perplexity

Power Demand Forecasting Issues: Floor Overestimation and Model Retraining

Based on extensive research into power demand forecasting systematic errors and model performance degradation, I can provide comprehensive insights into your forecasting challenges.

Primary Causes of Floor Overestimation

The systematic overestimation of demand floors (minimum demand levels) in your power forecasting model is a well-documented phenomenon with several key causes:

1. Model Bias and Systematic Errors

Persistence Forecast Effect: Research shows that when forecasting models lack sufficient statistical regularity in historical data, predictions tend to regress towards the most recently observed consumption values[1]. This creates a systematic bias where minimum demand levels are consistently overestimated.

Structural Bias in Model Training: Studies demonstrate that forecasting models can develop systematic biases due to the training process itself. When models are trained on historical data that includes periods of higher baseline consumption, they tend to maintain these elevated baseline expectations even when actual demand patterns shift lower[2][3].

Weather-Demand Relationship Bias: Research indicates that models often struggle with the non-linear relationship between weather conditions and minimum demand periods. During mild weather conditions, the temperature-demand relationship weakens, but models may still apply learned temperature coefficients inappropriately[4].

2. Concept Drift and Model Degradation

Concept Drift Impact: Studies show that power demand forecasting models experience significant performance degradation over time due to concept drift, where the underlying data distribution changes[5][6]. This degradation can range from 5-15% in accuracy and specifically affects minimum demand prediction accuracy.

Seasonal Pattern Drift: Research reveals that seasonal changes influence the scale, mean, and median values in input data, changing their distribution and causing models to systematically overestimate during low-demand periods[7].

3. Feature Engineering Limitations

Insufficient Low-Demand Feature Representation: Academic research suggests that traditional feature engineering approaches often fail to capture the specific characteristics of minimum demand periods[8][9]. Your current feature set may be optimized for peak demand prediction rather than floor-level accuracy.

Missing Temporal Dependencies: Studies indicate that minimum demand forecasting requires different temporal features than peak demand forecasting. The lag features and rolling windows that work well for peak periods may not effectively capture the patterns in minimum demand scenarios[10].

Model Retraining: Fresh Data vs. Extended Dataset

Recent research provides clear guidance on this critical decision:

Evidence Supporting Fresh Data Approach

Reduced Computational Costs: Studies demonstrate that less frequent retraining strategies maintain forecast accuracy while reducing computational costs by up to 90%[11][12]. For power demand forecasting, retraining every 2-4 weeks often provides optimal balance between accuracy and efficiency.

Concept Drift Management: Research shows that models trained on fresh, recent data adapt better to current demand patterns than those trained on extended historical datasets[13][14]. This is particularly important for minimum demand forecasting where consumption patterns can shift more dramatically.

Performance Stability: Studies indicate that daily retraining (continuous updating) doesn't necessarily improve accuracy and can actually introduce noise. Weekly or bi-weekly retraining schedules often provide better performance[15][11].

When Extended Datasets Are Beneficial

Seasonal Pattern Learning: Research suggests that extended datasets are valuable when your forecasting needs to capture long-term seasonal patterns that occur over multiple years[16]. However, this benefit must be balanced against concept drift risks.

Structural Change Detection: Studies show that maintaining longer historical datasets helps in detecting structural changes in demand patterns, but the optimal length is typically 2-3 years rather than indefinite extension[16].

Recommended Solutions

1. Bias Correction Techniques

Implement Systematic Bias Correction: Research demonstrates that two-stage correction methods (bias calibration and error propagation adjustment) can nearly eliminate forecast bias in power demand forecasting[17][18].

Weather Bias Correction: Studies show that implementing temperature bias correction specifically for minimum demand periods can significantly improve floor-level predictions[4][19].

2. Enhanced Feature Engineering

Minimum Demand Specific Features: Research suggests implementing features specifically designed for minimum demand prediction, including:

- Time-of-day specific lag features for off-peak periods
- Weather interaction features optimized for low-demand scenarios
- Economic indicators that affect baseline consumption[10][8]

Decomposition-Based Features: Studies recommend using signal decomposition techniques (EMD, VMD) to separate trend, seasonal, and residual components, which can improve minimum demand accuracy[20][21].

3. Model Architecture Improvements

Ensemble Methods with Bias Correction: Research shows that combining multiple models with specific bias correction for minimum demand periods can reduce systematic errors[22][20].

Quantile Regression: Studies demonstrate that implementing quantile regression specifically for minimum demand prediction can better capture floor-level patterns[23][6].

4. Optimal Retraining Strategy

Implement Adaptive Retraining: Research suggests using a concept drift detection system that triggers retraining only when systematic changes are detected, rather than following fixed schedules[6][24].

Fresh Data Window: Studies recommend using a rolling window of 1-2 years of recent data for retraining, balancing between capturing relevant patterns and avoiding outdated information[16][11].

Evaluation and Monitoring

Bias-Specific Metrics: Research recommends implementing evaluation metrics that specifically track over-forecasting and under-forecasting bias, particularly for minimum demand periods[25][17].

Performance Monitoring: Studies suggest continuous monitoring of forecast bias using techniques like the Continuous Ranked Probability Score for probabilistic forecasts[6].

Conclusion

Your observation about floor overestimation reflects a common systematic bias in power demand forecasting. The research strongly suggests that **retraining with fresh data on a weekly or bi-weekly schedule**, combined with **bias correction techniques** and **minimum demand-specific feature engineering**, will likely provide better results than extending your historical dataset indefinitely.

The key is implementing a systematic approach that addresses the specific characteristics of minimum demand periods while maintaining computational efficiency through optimal retraining schedules.